



# High-Order ALE Hydrodynamics with Tangential Movement on Curved Boundaries

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23 September 2026

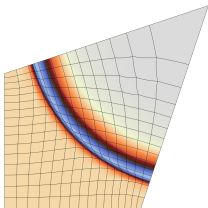
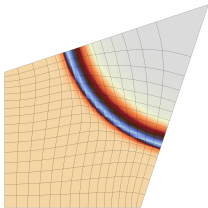
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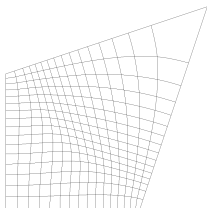
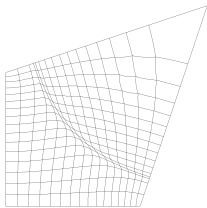
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# Arbitrary Lagrangian-Eulerian (ALE) Methods

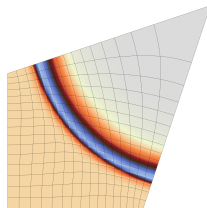
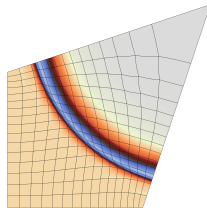
**Lagrangian Phase**



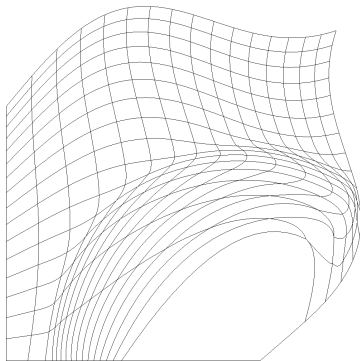
**Remesh Phase**



**Remap Phase**



# Goal: Implement an ALE Method on Domains with Curved Boundaries



## Slip boundary conditions

$$\mathbf{v} \cdot \mathbf{n} = 0 \quad \text{but} \quad \mathbf{v} \cdot \boldsymbol{\tau} \text{ is free.}$$

- ▶ Boundary nodes should remain on the boundary during the Lagrangian and remesh phases.
- ▶ Boundary conditions are enforced weakly during the Lagrangian phase.
- ▶ A parameterization of the boundary allows boundary nodes to move tangentially during the remesh.

# 1. Lagrangian Phase: Move Mesh with the Fluid

Non-conservative Euler equations:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad \rho J = \rho_0,$$

$$\rho \frac{d\mathbf{v}}{dt} = \nabla \cdot \boldsymbol{\sigma}, \quad \rho \frac{de}{dt} = \nabla \mathbf{v} : \boldsymbol{\sigma}.$$

$$\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\sigma}_a, \quad p = \text{EOS}(\rho, e).$$

$\mathbf{x}$  : material position

$\rho$  : density

$J = \det(\nabla_{\mathbf{x}_0} \mathbf{x})$  : deformation Jacobian

$\boldsymbol{\sigma}$  : Cauchy stress

EOS : equation of state

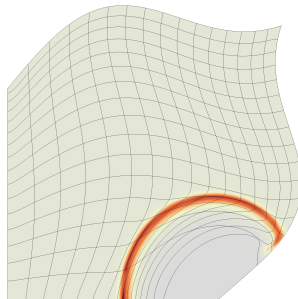
$\mathbf{v}$  : velocity

$\rho_0$  : initial density

$e$  : specific internal energy

$p$  : pressure

$\boldsymbol{\sigma}_a$  : artificial viscosity



**High-order  $Q_k Q_{k-1}$  discretization:** On a quadrilateral or hexahedral mesh,  $\mathbf{x}, \mathbf{v} \in \mathcal{V} = (Q_k)^d \subset (H^1(\Omega_0))^d$  use a continuous Gauss–Lobatto basis  $\{w_i\}$  and  $e \in \mathcal{E} = \hat{Q}_{k-1} \subset L^2(\Omega_0)$  uses a discontinuous Bernstein basis  $\{\phi_j\}$ .

# Weak Slip-Wall Enforcement

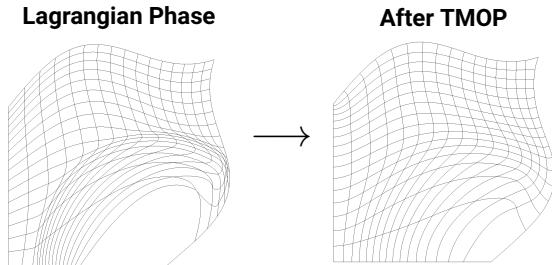
$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \mathbf{v}, & \mathbf{M}_V \frac{d\mathbf{v}}{dt} &= -\mathbf{F}\mathbf{1}, & \mathbf{M}_E \frac{d\mathbf{e}}{dt} &= \mathbf{F}^T \mathbf{v}. \\ (\mathbf{M}_V)_{i\ell jm} &= \int_{\Omega} \rho \mathbf{w}_i \mathbf{w}_j \delta_{\ell m} + \int_{\Gamma_0} \alpha_0 \rho_{\max} L \mathbf{w}_i \mathbf{n}_{0,\ell} \mathbf{w}_j \mathbf{n}_{0,m}, & (\mathbf{M}_E)_{ij} &= \int_{\Omega} \rho \phi_i \phi_j. \\ (\mathbf{F})_{i\ell j} &= \int_{\Omega} \sigma_{\ell m} \phi_j \nabla_{x_m} \mathbf{w}_i - \int_{\Gamma} (n_m \sigma_{mk} n_k) n_{\ell} \mathbf{w}_i \phi_j + \int_{\Gamma} \beta \rho c_s (\mathbf{v} \cdot \mathbf{n}) \mathbf{w}_i n_{\ell} \phi_j. \end{aligned}$$

- ▶ The boundary mass penalty term  $\alpha_0 \rho_{\max} L \mathbf{n}_0 \otimes \mathbf{n}_0$  suppresses normal acceleration.
- ▶ The normal-projected term  $(\mathbf{n}^T \boldsymbol{\sigma} \mathbf{n}) \mathbf{n}$  eliminates tangential traction while retaining normal traction.
- ▶ The acoustic-impedance term  $\beta \rho c_s (\mathbf{v} \cdot \mathbf{n}) \mathbf{n}$  damps residual normal velocity.

$$\begin{aligned} \alpha_0 &= \beta L / \det(\nabla_{\hat{\mathbf{x}}} \mathbf{x}_0)^{1/d} : \text{mass-penalty scale} & \beta &= \lambda C_I : \text{impedance coefficient} \\ C_I &: (k+1)^2 & c_s &: \text{sound speed } (\sqrt{\gamma(\gamma-1)}e \text{ for an ideal gas}) \\ L &: \text{characteristic length} & \mathbf{n}_0, \mathbf{n} &: \text{reference and current wall normals} \end{aligned}$$

*"Weak boundary conditions for Lagrangian shock hydrodynamics: a high-order finite element implementation on curved boundaries," Atallah, Tomov, Scovazzi, 2023.*

## 2. Remesh Phase: Improve Mesh Quality



- ▶ The Target-Mesh Optimization Paradigm (TMOP) moves high-order mesh nodes to minimize a global target-matrix quality objective.
- ▶  $T = AW^{-1}$  maps the target to the physical element. Newton minimizes  $F = \sum_E \int [\omega \mu(T) + \xi(\mathbf{x} - \mathbf{x}_0, \delta)]$ :  $\mu$  controls skew and aspect ratio, while  $\xi$  limits displacement from the Lagrange mesh.
- ▶ At a curved wall,  $\mathbf{x}_i = S(\mathbf{p}_i)$ : Newton updates one curve parameter or two surface parameters instead of boundary Cartesian coordinates.

*"The target-matrix optimization paradigm for high-order meshes," Dobrev, Knupp, Mittal, Tomov, 2019.*

## TMOP: Parameterized Boundary Motion

- ▶ Parameterize boundary degrees of freedom by  $\mathbf{p}_i = t_i$  on curves and edges, or  $\mathbf{p}_i = (u_i, v_i)$  on 3D faces, then recover  $\mathbf{x}_i = S(\mathbf{p}_i)$ .
- ▶ Solve one Newton system with Cartesian interior variables and parameterized boundary variables.
- ▶ Apply the chain rule to form the boundary-node derivatives.

$$\frac{\partial A_{ab}}{\partial p_{ri}} = \frac{\partial S_a}{\partial p_r}(\mathbf{p}_i) \frac{\partial w_i}{\partial \bar{x}_b},$$

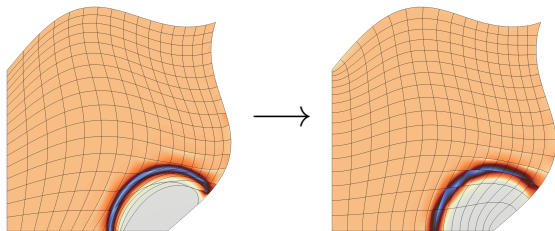
$$\frac{\partial^2 A_{ab}}{\partial p_{si} \partial p_{ri}} = \frac{\partial^2 S_a}{\partial p_s \partial p_r}(\mathbf{p}_i) \frac{\partial w_i}{\partial \bar{x}_b}.$$

- ▶ Store the post-Lagrangian normal offset  $d_i$  and update only the boundary parameter.

$$\mathbf{x}_i^{\text{new}} = S(\mathbf{p}_i + \Delta \mathbf{p}_i) + d_i \mathbf{n}(\mathbf{p}_i + \Delta \mathbf{p}_i).$$

- ▶ If  $d_i = 0$ , the node remains exactly on  $S$ ; otherwise it follows an offset curve or surface.
- ▶ Keeping  $d_i$  fixed avoids projection-induced element inversion and makes the remesh motion tangential for remap.

### 3. Remap Phase: Transfer the State Conservatively



Over pseudo-time  $\tau \in [0, 1]$ ,

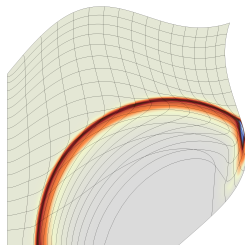
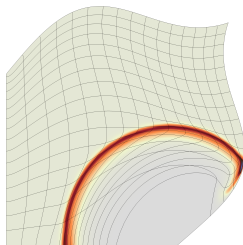
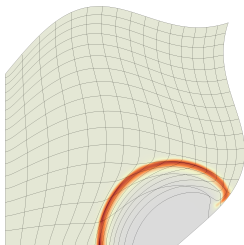
$$\mathbf{x}(\tau) = \mathbf{x}^{\text{old}} + \tau \mathbf{u}, \quad \mathbf{u} = \mathbf{x}^{\text{new}} - \mathbf{x}^{\text{old}}.$$

$$M_q \frac{d\mathbf{q}}{d\tau} = K_q \mathbf{q}, \quad \mathbf{q} \in \{\rho, \mathbf{e}, \mathbf{v}\}.$$

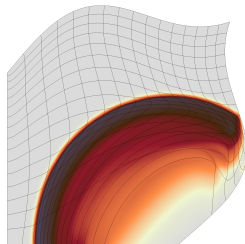
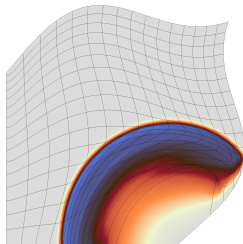
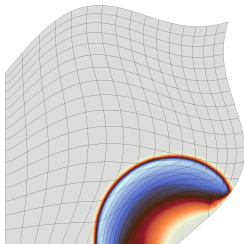
- $K_q$  uses conservative upwind fluxes; density and energy use FCT limiting, while velocity uses a limited  $H^1$  method.

# Curved Square

Density

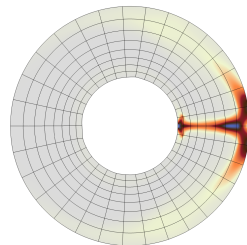
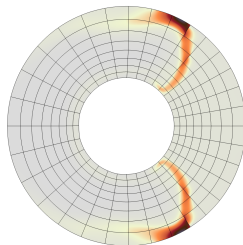
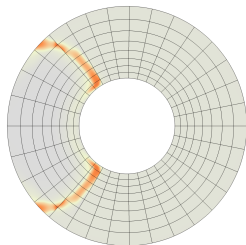


Velocity

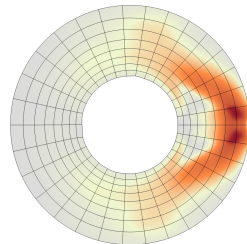
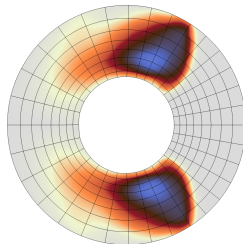
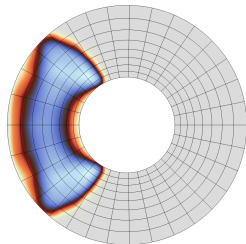


# Annulus

Density

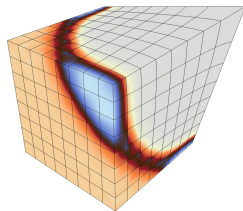
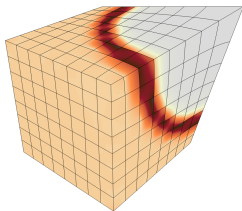


Velocity

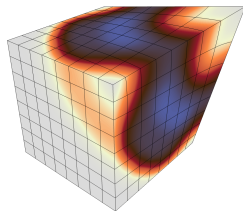
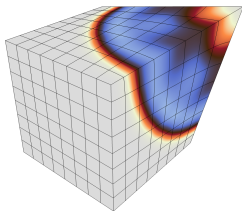


# Stretched Cube

Density

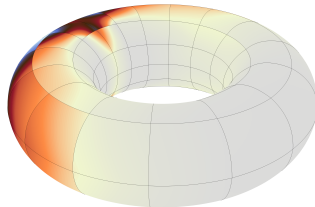
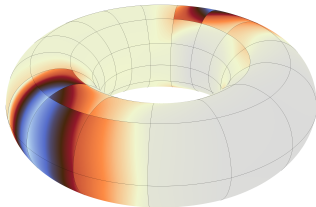


Velocity

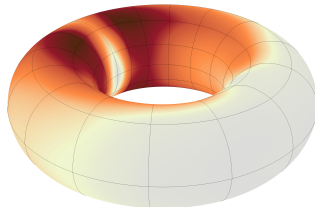
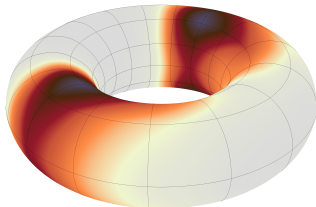


# Solid Torus

Density



Velocity



## Takeaways

- ▶ Weak slip-wall enforcement permits tangential motion in the Lagrangian phase.
- ▶ Parameterized TMOP preserves curved boundaries while repairing the mesh.
- ▶ Tangential remeshing makes conservative remap compatible with the wall.



